

Pharmacology Study Guide (Chapter 11)

Focused Review for NBDHE Preparation (Questions 1 to 30)

How to use this guide: Master the yellow "High-Yield" callout boxes first. They contain the core pharmacology facts tested directly in Questions 1 to 30 on the NBDHE Dental Hygiene Board Exam. Then utilize the question-to-concept map and clinical vocabulary table to cement your mastery.

1. Autonomic & Cardiovascular Pharmacology

- **Angiotensin-Converting Enzyme (ACE) Inhibitors:**
 - **Prototype Drugs:** Enalapril (Vasotec), Lisinopril, Captopril (suffix "-pril").
 - **Mechanism of Action:** Inhibit conversion of Angiotensin I to Angiotensin II (a potent vasoconstrictor); reduce aldosterone secretion.
 - **Adverse Effects & Dental Significance:** Dry, nonproductive hacking cough (due to accumulated bradykinin in 10–20% of patients); angioedema; orthostatic hypotension.
- **Calcium Channel Blockers (CCBs):**
 - **Prototype Drugs:** Nifedipine (Procardia), Amlodipine (Norvasc), Diltiazem, Verapamil (suffix "-dipine").
 - **Mechanism:** Inhibit calcium ion influx into vascular smooth muscle and myocardium, causing vasodilation.
 - **High-Yield Oral Complication: Drug-Induced Gingival Enlargement (Hyperplasia)** in 15–80% of patients. Begins in interdental papillae.
- **Beta-Adrenergic Receptor Blockers:**
 - **Prototype Drugs:** Metoprolol (Lopressor, β_1 -selective), Propranolol (Inderal, non-selective) (suffix "-olol").
 - **Adverse Effects:** Bradycardia, bronchospasm (with non-selective agents). *Epinephrine Interaction:* Limit local anesthetic epinephrine with nonselective beta-blockers (cardiac dose: max 0.04 mg epinephrine).
- **Antihyperlipidemics (Lipid-Lowering Drugs):**
 - **HMG-CoA Reductase Inhibitors ("Statins"):** Lovastatin (Mevacor), Atorvastatin (Lipitor). Block rate-limiting step in cholesterol synthesis.
 - **Fibric Acid Derivatives (Fibrates):** Fenofibrate (Tricor), Gemfibrozil. Lower triglycerides and VLDL; do **NOT** alter blood coagulation or clotting times.
 - **Grapefruit Juice Interaction:** Inactivates intestinal CYP3A4 enzymes, dramatically increasing serum levels and toxicities of statins and calcium channel blockers.

HIGH-YIELD FOR QUESTIONS 6, 8, 10, 11, 15, 17, 18:

- * Enalapril is an **ACE inhibitor** (associated with chronic dry cough).
- * Nifedipine is a **calcium channel blocker** that causes **gingival hyperplasia**.
- * Metoprolol is a **cardioselective beta-blocker**.
- * Fenofibrate treats elevated lipids and **does NOT affect blood coagulation**.
- * Grapefruit juice inhibits **CYP3A4**, causing severe drug-food toxicity interactions.

2. Central Nervous System & Psychiatric Medications

- **Trigeminal Neuralgia (*Tic Douloureux*):**
 - **Drug of Choice: Carbamazepine (Tegretol)** is the first-line anticonvulsant for severe neuropathic lancinating facial pain.
 - **Pediatric Alert:** Chewable carbamazepine contains **63% sucrose**, putting pediatric patients at high risk for rampant dental caries.
- **Antidepressants:**
 - **Selective Serotonin Reuptake Inhibitors (SSRIs):** Fluoxetine (Prozac), Sertraline (Zoloft). Solely block serotonin reuptake (SERT); low anticholinergic liability.
 - **Tricyclic Antidepressants (TCAs):** Amitriptyline (Elavil), Nortriptyline. Potent anticholinergic side effects (**severe xerostomia**).
- **Antipsychotics (Neuroleptics):**

- **First-Generation (Typical):** Fluphenazine, Thioridazine (phenothiazines). Potent D2 dopamine antagonists causing Extrapyramidal Symptoms (EPS: acute dystonia, akathisia, parkinsonism) and marked dry mouth.
- **Second-Generation (Atypical):** Risperidone (Risperdal), Aripiprazole (Abilify). Dual 5-HT_{2A} and D2 antagonists with lower EPS risks.
- **Oral Effect:** Antipsychotics cause **xerostomia** due to muscarinic blockade; **sialorrhea (excess salivation) is the exception**.
- **Anxiolytics & Hypnotics (Sedation):**
- **Alprazolam (Xanax):** Short-to-intermediate benzodiazepine; first-line oral premedication for **acute dental anxiety**.
- **Zolpidem (Ambien):** Non-benzodiazepine GABA-A agonist hypnotic; associated with complex parasomnias (**sleep-driving, sleep-eating**).

HIGH-YIELD FOR QUESTIONS 1, 2, 4, 9, 13, 14, 21, 23, 30:

- * **Carbamazepine** treats **trigeminal neuralgia**; chewable tablets contain **63% sugar**.
- * **Fluoxetine** is an SSRI that solely blocks serotonin reuptake with minimal xerostomia.
- * **Thioridazine** is a first-generation phenothiazine antipsychotic causing dry mouth.
- * **Antipsychotics** cause EPS and dry mouth; **sialorrhea is NOT a typical side effect**.
- * **Alprazolam** is used for **acute situational dental anxiety**.
- * **Zolpidem** causes **sleep-driving and sleep-eating**.

3. Analgesics, Anti-Inflammatories & Emergency Reversal

- **Aspirin (Acetylsalicylic Acid):**
- **Mechanism:** Irreversibly inhibits **COX-1**, suppressing platelet Thromboxane A₂ (TXA₂) synthesis.
- **Clinical Dental Effect:** Prolongs bleeding time for the entire **7 to 10 day lifespan of the platelet**, causing **increased bleeding during scaling and root planing**.
- **Nonsteroidal Anti-Inflammatory Drugs (NSAIDs):**
- **Examples:** Ibuprofen, Naproxen, Meloxicam.
- **Mechanism:** Inhibit renal vasodilatory prostaglandins (PGE₂, PGI₂), causing renal vasoconstriction and decreased GFR. **Avoid in renal impairment**.
- **Acetaminophen (Tylenol / APAP):**
- **Safety Profile:** Safe analgesic of choice in **renal impairment** and pregnancy (Category B).
- **Toxicity:** Overdose (>3000–4000 mg/day) depletes glutathione, generating toxic metabolite **NAPQI**, causing fatal **acute liver toxicity (hepatotoxicity)**.
- **Opioids & Overdose Reversal:**
- **Naloxone (Narcan):** Pure competitive **opioid receptor antagonist** that rapidly reverses life-threatening opioid-induced respiratory depression and coma.

HIGH-YIELD FOR QUESTIONS 16, 25, 26, 27:

- * **Aspirin** causes **irreversible platelet inhibition** and increases bleeding during SRP.
- * **Acetaminophen** is safe in **renal failure**, but toxic doses cause **severe liver necrosis (NAPQI)**.
- * **Naloxone (Narcan)** is the emergency drug of choice to reverse **opioid overdose**.

4. Antibiotics, Pregnancy Safety & Gastrointestinal Agents

- **Tetracyclines (Doxycycline, Minocycline):**
- **Contraindication in Pregnancy:** Category D. Chelates with calcium orthophosphate, depositing into calcifying fetal bones and unerupted tooth germs, causing **permanent brownish-yellow intrinsic staining** and enamel hypoplasia.
- **Photosensitivity:** Causes exaggerated sunburn reactions.
- **Safe Antibiotics in Pregnancy:**
- **Amoxicillin & Penicillin VK:** FDA Category B; safe for odontogenic infections during pregnancy.
- **Metronidazole (Flagyl) & Alcohol Warning:**
- Inhibits aldehyde dehydrogenase, causing a **severe disulfiram (Antabuse) reaction** (nausea, violent vomiting, flushing) with any alcohol, including **alcohol-containing mouth rinses**.

- **Gastrointestinal Drugs:**
- **Proton Pump Inhibitors (PPIs):** Esomeprazole (Nexium), Omeprazole. **Irreversibly bind** H⁺/K⁺ ATPase pump (>24 hr acid suppression).
- **H₂-Receptor Antagonists:** Famotidine (Pepcid). Competitively block gastric H₂ receptors.
- **Peptic Ulcer Triple Therapy:** PPI + Clarithromycin + Amoxicillin (or Metronidazole) to eradicate *Helicobacter pylori*.

HIGH-YIELD FOR QUESTIONS 3, 5, 7, 12, 20, 22, 24:

- * **Tetracycline** is contraindicated in pregnancy (fetal tooth staining) and causes **photosensitivity**.
- * **Amoxicillin** is **safe for pregnant patients**.
- * Patients taking **metronidazole** must strictly avoid **alcohol-based mouth rinses**.
- * **Esomeprazole** produces **>24-hour acid suppression** via irreversible proton pump binding.
- * **Peptic ulcer disease (H. pylori)** is treated with **clarithromycin + amoxicillin/metronidazole + PPI**.

5. Respiratory Drugs & Smoking Cessation

- **Asthma Management:**
- **Rescue Inhaler (Acute Attack):** **Albuterol** (Short-Acting Beta-2 Agonist - SABA); rapid onset bronchodilation.
- **Maintenance Therapy:** Salmeterol (LABA), Fluticasone / Fluticasone (inhaled corticosteroid; instruct patient to rinse mouth to prevent oral candidiasis).
- **Smoking Cessation Pharmacotherapy:**
- **Nicotine Gum:** Proper "chew and park" technique prevents jaw ache and nausea. Side effects: mouth soreness, sticking to dental work. **Runny nose is an EXCEPTION (side effect of nasal spray)**.
- **Varenicline (Chantix):** $\alpha 4\beta 2$ nicotinic partial agonist. Side effects: vivid dreams, mood changes, nausea, **constipation** (diarrhea is an exception).

HIGH-YIELD FOR QUESTIONS 19, 28, 29:

- * **Albuterol** is the primary rescue inhaler for **acute asthma symptoms**.
- * **Nicotine gum** causes jaw ache and sticks to dental work; **runny nose occurs with nasal spray**.
- * **Varenicline (Chantix)** causes abnormal dreams, mood changes, and **constipation**.

6. Bilingual Pharmacology Master Terminology Table

| English Dental Term | Spanish Dental Translation | Clinical & Board Relevance |

| :--- | :--- | :--- |

- | **Trigeminal neuralgia** | *Neuralgia del trigémino* | Neuropathic lancinating facial pain treated with carbamazepine. |
- | **Xerostomia (Dry mouth)** | *Xerostomía (Boca seca)* | Caused by TCAs, phenothiazines, antihistamines, and diuretics. |
- | **Gingival hyperplasia** | *Hiperplasia gingival* | Drug-induced enlargement caused by CCBs, phenytoin, and cyclosporine. |
- | **Irreversible platelet inhibition** | *Inhibición plaquetaria irreversible* | Caused by aspirin (COX-1 block); lasts 7–10 days. |
- | **Hepatotoxicity** | *Hepatotoxicidad* | Acute liver damage caused by acetaminophen metabolite NAPQI. |
- | **Disulfiram-like reaction** | *Reacción tipo disulfiram* | Violent nausea from combining metronidazole with alcohol. |
- | **Proton pump inhibitor (PPI)** | *Inhibidor de la bomba de protones* | Irreversible gastric acid suppression lasting >24 hours (esomeprazole). |
- | **Photosensitivity** | *Fotosensibilidad* | Exaggerated sunburn risk caused by tetracyclines. |
- | **Extrapyramidal symptoms (EPS)** | *Síntomas extrapiramidales* | Motor dystonias and tremors caused by D₂-blocking antipsychotics. |
- | **Rescue inhaler** | *Inhalador de rescate* | Short-acting beta-2 agonist (albuterol) for acute bronchospasm. |